

Foundation of Blockchain Technology (IS501) Pre-Reading Material – Lecture 10

Blockchain Consensus Challenges & Solution - A

Estimated Reading Time: 15–20 minutes

Why This Topic Matters

Blockchain networks require participants to verify information before accepting it. However, traditional verification may require a participant to reveal sensitive information.

Zero-Knowledge Proofs (ZKPs) provide a way to prove that a statement is true **without revealing the secret information behind that statement**.

This concept is particularly important for privacy-preserving blockchain applications, where participants need to demonstrate that they have performed a computation correctly without exposing their private data or computational details.

In learning-based blockchain systems, ZKPs can be used to prove the correctness or performance of a machine-learning model without directly revealing the model parameters.

Learning Outcomes

After completing this pre-reading, you should be able to:

- Explain the basic concept of Zero-Knowledge Proofs.
- Identify the roles of a prover and verifier.
- Explain why privacy-preserving verification is important in blockchain.
- Explain the meaning of zk-SNARK.
- Describe the basic working of zk-SNARK.
- Explain the **Setup, Prove, and Verify** algorithms.
- Understand completeness, succinctness, proof of knowledge, and zero knowledge.
- Explain why zk-SNARKs are useful for blockchain applications.



Activate Your Prior Knowledge

1. Proving Without Revealing

Suppose Alice says:

“I know the password to this account.”

Bob wants to verify Alice's claim.

The simplest solution is for Alice to reveal the password.

Think: Is revealing the password a good verification mechanism?

2. Blockchain Verification

In blockchain, every participant may need to verify whether a transaction or computation is valid.

Think: What happens if verification requires revealing private information?

3. Machine-Learning Model

Suppose a participant claims:

“My machine-learning model achieves 95% accuracy.”

Other blockchain nodes want to verify this claim.

Think: Does the participant necessarily need to reveal the complete model to prove the claim?

Prerequisite Concepts

You should be familiar with:

- Blockchain

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Course Repository: <https://github.com/rajesh101191/Foundation-of-Blockchain-Technology>



- Blockchain transactions
- Consensus mechanisms
- Cryptographic hashing
- Public and private keys
- Digital signatures
- Basic cryptography
- Smart contracts
- Basic machine-learning concepts

Core Concepts

1. Zero-Knowledge Proof

A **Zero-Knowledge Proof (ZKP)** is a cryptographic technique that allows one party, called the **prover**, to convince another party, called the **verifier**, that a statement is true without revealing the underlying secret information.

Simple Example

Suppose:

Statement:

“I know a secret number that satisfies a particular mathematical condition.”

The prover should be able to convince the verifier that the statement is true without revealing the secret number itself.

Key Idea

Prove knowledge without revealing knowledge.

2. Prover

The **prover** is the participant who possesses secret information and wants to prove that a particular statement is true.

For example:

- A private key
- A password

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- Private data
- Model parameters
- A secret computation

3. Verifier

The **verifier** is the participant who checks whether the proof generated by the prover is valid.

The verifier should be convinced that the statement is true without learning the prover's secret information.

4. Witness

The secret information possessed by the prover is generally called the **witness**.

For example:

Statement	Possible Witness
I know the password	Password
I know the private key	Private key
My model achieves 95% accuracy	Model parameters
I performed the computation correctly	Private computation/data

5. Why Do We Need ZKP in Blockchain?

Blockchain emphasizes transparency and verification.

However, complete transparency can conflict with privacy.

Consider:

Participant → Model → Blockchain Network

If every participant receives the complete model, the model may be exposed.

Instead, we want:

Participant → Proof → Blockchain Network

The blockchain participants verify the proof without directly obtaining the secret model.

This creates an important principle:

Verification does not necessarily require disclosure.

6. zk-SNARK

One important type of Zero-Knowledge Proof is called:

zk-SNARK

Zero-Knowledge Succinct Non-Interactive Argument of Knowledge

The term can be understood as follows:

Zero-Knowledge

The secret information is not revealed.

Succinct

The proof is relatively compact.

Non-Interactive

The proof can be generated and verified without repeated interaction between prover and verifier.

Argument

The validity is based on computational assumptions.

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Knowledge

A valid proof demonstrates that the prover possesses the required information.

7. Basic zk-SNARK Workflow

A simplified zk-SNARK system consists of three major stages:

Setup

The system generates:

- Proving Key (**pk**)
- Verification Key (**vk**)

Prove

The prover uses:

- Proving key
- Public input
- Private witness

to generate a proof:

$$\pi = \text{Prove}(\text{pk}, x, w)$$

Verify

The verifier uses:

- Verification key
- Public input
- Proof

to determine whether the proof is valid.

Verify(vk, x, π) $\rightarrow$ 0 or 1

where:

1 = Valid

0 = Invalid

8. Completeness

Completeness means that if the statement is true and the prover is honest, the verifier should accept the proof.

In simple words:

A correct proof should be accepted.

9. Succinctness

A zk-SNARK proof should be relatively small compared with the computation being proved.

This is important in blockchain because:

- Storage is limited.
- Communication takes resources.
- Many nodes may need to verify the proof.

10. Proof of Knowledge

A valid proof should demonstrate that the prover actually possesses the required witness.

In simple words:

A participant should not be able to generate a valid proof without actually possessing the required secret information.



11. Zero Knowledge

The verifier learns that the statement is true, but does not learn the underlying secret information.

For example:

Without ZKP

Model → Verifier

With ZKP

Model → Proof → Verifier

The verifier receives the proof rather than the complete secret model.

12. ZKP and Machine Learning

Consider a machine-learning participant.

The participant trains a model using private data.

The participant claims:

Accuracy = 95%

Other blockchain participants need to verify the claim.

Traditional approach:

Model → Other Nodes → Testing → Accuracy Verification

Problem:

Model exposure → Privacy risk

ZKP-based approach:

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Private Model → ZK Proof → Blockchain Nodes

The nodes verify the proof without directly receiving the private model.

This idea forms the foundation for the **Zero-Knowledge Proof of Training (ZKPoT)** approach.

Guided Reading Questions

1. What is a Zero-Knowledge Proof?
2. Who is the prover?
3. Who is the verifier?
4. What is a witness?
5. Why is privacy important in blockchain?
6. What does zk-SNARK stand for?
7. What are the three major stages of a zk-SNARK?
8. What is completeness?
9. What is succinctness?
10. What is meant by proof of knowledge?
11. Why is zero knowledge useful for blockchain?
12. How can ZKP be used to verify a machine-learning model?

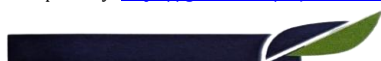
Reflection Activity

Write **100–150 words** explaining:

Why might it be useful for a blockchain participant to prove that a computation was performed correctly without revealing the underlying data or computation?

Your answer should discuss:

- Verification
- Privacy
- Blockchain
- Zero-Knowledge Proofs



Self-Assessment

Fill in the blanks:

1. A participant who generates a ZKP is called the _____.
2. The participant who checks the proof is called the _____.
3. The secret information used by the prover is called the _____.
4. ZKP stands for _____.
5. zk-SNARK stands for _____.
6. The three basic zk-SNARK stages are Setup, _____, and Verify.
7. The verification key is generally represented by _____.
8. The proof generated by the prover is represented by _____.
9. The property that prevents disclosure of the secret is called _____.
10. A compact proof is related to the property of _____.

Key Takeaway

A **Zero-Knowledge Proof** allows a participant to prove that a statement is true **without revealing the secret information used to establish that statement**.

In blockchain, this provides an important combination:

Verification + Privacy

The next lecture will build on this foundation and examine how Zero-Knowledge Proofs can be integrated directly into a blockchain **consensus mechanism**.

